# Lecture VII - Library Routing Optimization

Applied Optimization with Julia

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## Introduction

### Learning Objectives

By the end of this lecture, you will be able to:

- **Formulate** a vehicle routing problem as a mathematical model
- **Explain** how subtours arise and how to prevent them
- **Assess** why large instances are hard to solve exactly
- **Choose** between exact methods and heuristics in practice

## Central Libraries

**Question:** Does anyone have an idea what a central library is?

### Central Libraries

- **Book Delivery to Libraries** in Germany
- They supply **all local libraries** within the same state
- Complex, as **number of libraries** per state can be large
- Books and media in the libraries **change often**
- Customers can **request books** from other libraries

### Structure of the Deliveries

- For delivery, the central library has **several employees and cars**
- Local libraries differ in size, some receive **more items**
- Items are **collected as well** during the tours[1]
- They are **transported back** to the central library

### Potential Decisions

**Question**: **What decisions can the library make for tours?**

- Subdivide set of libraries into **several ordered tours**
- Decide in **which order to visit** the libraries
- Evaluate **which car to use** for each of the tours

[1] Due to regulations, the delivery tours cannot exceed a certain duration

- Decide **which driver** to assign to each of the tours

### Impact of the Decisions

**Question**: **What is the impact of the decisions?**

- Longer driving **increases the footprint** of the deliveries
- Suboptimal tours can lead to **unnecessary costs**
- Fuel, personnel, and repairs **are increased**
- Unhappy customers due to **waiting times** on books

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Have you heard of

this problem before?

## Problem Structure

### Objective

**Question**: **What could be the objective for central libraries?**

- **Lowering costs** through improved tours
- Improvement of their **footprint** through shorter tours
- **Faster fulfillment** of the deliveries

### Modelling

**Question**: **What could we try to model?**

**Minimization** of the travel time while **supplying all libraries** in the state and respecting **the vehicle capacities and driving time restrictions.**

### Your Turn

Play: https://beyondsimulations.github.io/Applied-Optimization/games/tours.html

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Capacitated

Vehicle Routing

(CVRP)

### Vehicle Routing Problem

- CVRP is a **subproblem**

- Main problem is **Vehicle Routing Problem (VRP)**
- Problem class about **designing routes for vehicle fleets**

> **Note**
> There are many variants of the VRP! E.g., with time windows, periodic deliveries, allowing for pickups or deliveries, **and much more!**

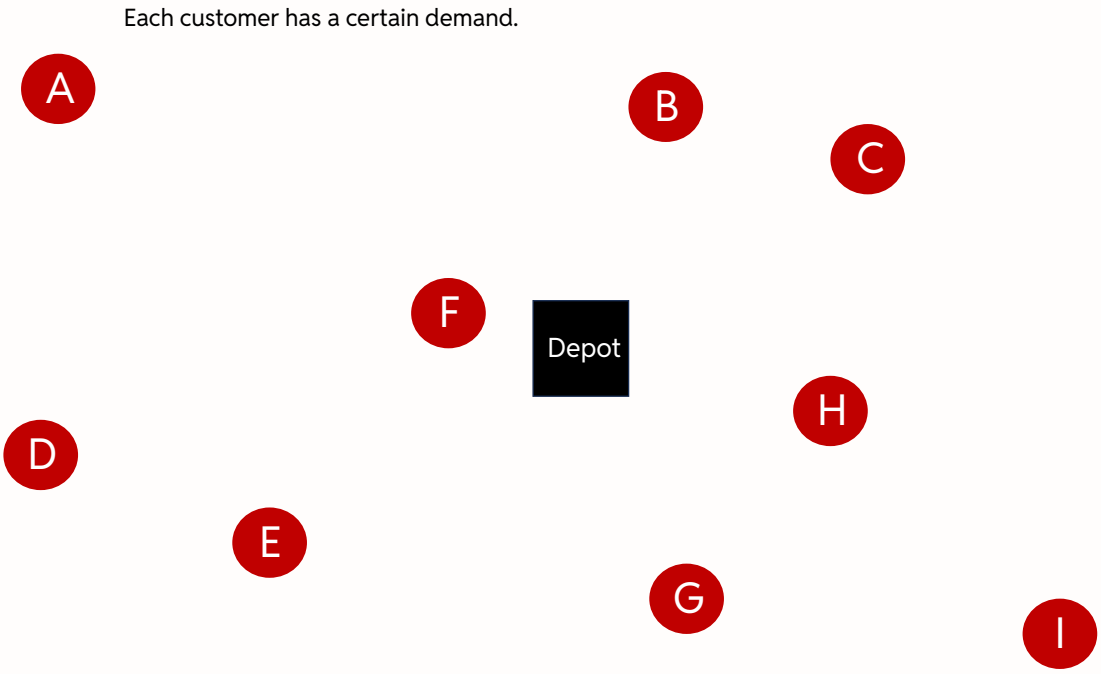
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Let's visualize

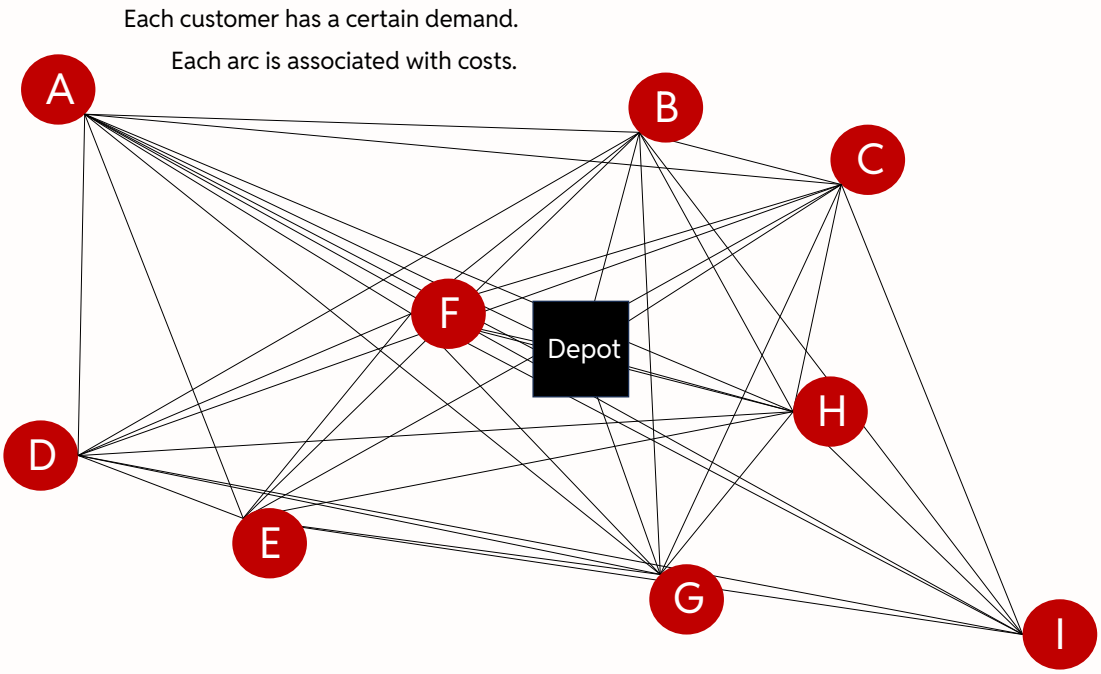
the problem!

# Problem Visualization

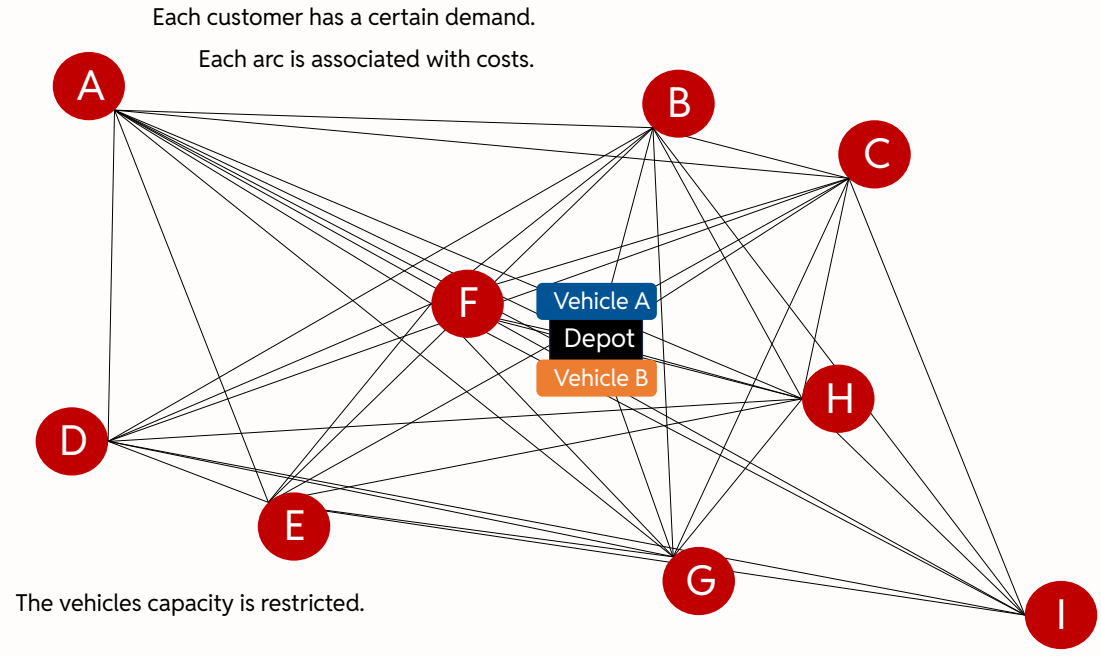
## Basic Problem Setting



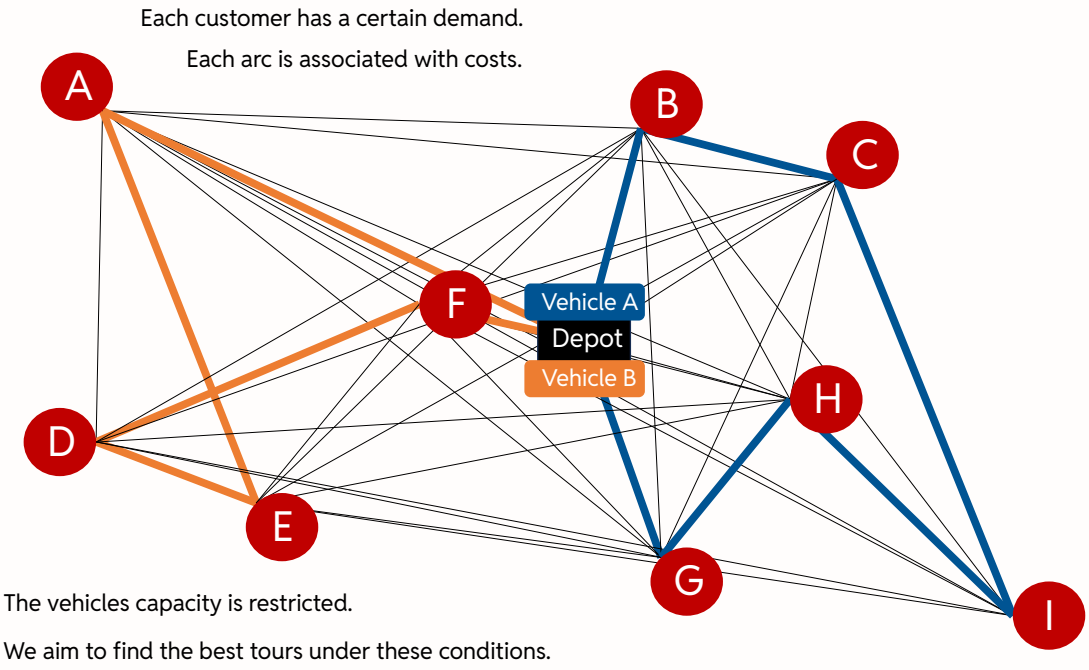
## Basic Setting with Arcs




## Setting with Vehicles



## Basic Setting with Tours




# Problem Structure

## Available Sets

**Question: What could be the sets here?**

- $\mathcal{V}$ - Set of all nodes, index $i \in \{0, 1, 2, ..., n\}$
- $\mathcal{A}$ - Set of all arcs between the nodes, index $(i, j) \in \mathcal{A}$
- $\mathcal{K}$ - Set of vehicles with identical capacity, index $k \in \mathcal{K}$
- $0 \in \mathcal{V}$ - Depot where the vehicles start

## Available Parameters

**Question: What are possible parameters?**

- $b$ - Capacity per vehicle
- $t$ - Maximal duration of each tour
- $d_i$ - Demand at node $i$
- $c_{i,j}$ - Travel time on an arc from $i$ to $j$

> **Tip**
> $t$ is the maximal duration of each tour, not the travel time on an arc or an index!

## Decision Variable(s)?

ℹ **We have the following sets:**
- All nodes, including the depot, $i \in \mathcal{V}$
- All arcs between the nodes, $(i, j) \in \mathcal{A}$
- The available vehicles, $k \in \mathcal{K}$

! **Our objective is to:**
Minimize the **total travel time** while supplying **all customers** and adhering to the **vehicle capacities and duration restrictions**.

**Question: What could be our decision variable/s?**

## Decision Variables

- $X_{i,j,k}$ - 1, if $k$ passes between $i$ and $j$ on its tour, 0 otherwise

ℹ **Variable Domain**
$X_{i,j,k}$ is a **binary variable**, as it can only take values 0 or 1. But most likely, you will already have spotted that!

**Question: Why might this make the problem difficult?**

## Decision Variable/s (again)?

ℹ **We have the following sets:**
- All nodes, including the depot, $i \in \mathcal{V}$
- All arcs between the nodes, $(i, j) \in \mathcal{A}$
- The available vehicles, $k \in \mathcal{K}$

! **Our objective is to:**
Minimization of the travel time (or driving distance), while supplying all customers and adhering to the vehicle capacities and duration restrictions. **Hint:** Even with many vehicles, **each arc can maximally be passed once**!

## Decision Variables (again)

**Question: What could be our decision variable/s?**

- $X_{i,j}$ - 1, if the arc between $i$ and $j$ is part of a tour, else 0

! **Only possible under certain conditions!**
Only possible, if time and capacity constraints are equal for all vehicles!

# Model Formulation

## Objective Function?

> ! **Our objective is to:**
> Minimization of the travel time (or driving distance), while supplying all customers and adhering to the vehicle capacities and duration restrictions.

**Question: What could be our objective function?**

> i **We need the following variable:**
> - $X_{i,j}$ - 1, if the arc between $i$ and $j$ is part of a tour, else 0

## Objective Function

> i **We need the following parameters:**
> - $c_{i,j}$ - travel time on an arc from $i$ to $j$

$$\text{Minimize} \quad \sum_{(i,j)\in\mathcal{A}} c_{i,j} \times X_{i,j}$$

**Question: What does $(i,j) \in \mathcal{A}$ under the sum mean?**

## From Math to Code

The objective function in **JuMP** looks very similar:

```
using JuMP, HiGHS

# Simple example with 4 nodes
nodes = ["depot", "A", "B", "C"]
arcs = [(i,j) for i in nodes, j in nodes if i != j]
c = Dict(("depot","A") => 10, ("depot","B") => 15, ("depot","C") => 20,
         ("A","depot") => 10, ("A","B") => 12, ("A","C") => 8,
         ("B","depot") => 15, ("B","A") => 12, ("B","C") => 5,
         ("C","depot") => 20, ("C","A") => 8, ("C","B") => 5)

model = Model(HiGHS.Optimizer)
@variable(model, X[arcs], Bin)
@objective(model, Min, sum(c[(i,j)] * X[(i,j)] for (i,j) in arcs))
```

> i **Note**
> The JuMP syntax `sum(c[(i,j)] * X[(i,j)] for (i,j) in arcs)` directly mirrors our mathematical notation $\sum_{(i,j)\in\mathcal{A}} c_{i,j} \times X_{i,j}$!

## Solving the Small Example

Let's add constraints and solve the example with **one vehicle**:

```
# Each node has to be entered and left exactly once
for n in nodes
    @constraint(model, sum(X[(i,j)] for (i,j) in arcs if j == n) == 1)
    @constraint(model, sum(X[(i,j)] for (i,j) in arcs if i == n) == 1)
end
set_silent(model)
optimize!(model)
println("Arcs: ", [a for a in arcs if value(X[a]) > 0.5])
println("Cost: ", objective_value(model))
```

```
Arcs: [("A", "depot"), ("depot", "A"), ("C", "B"), ("B", "C")]
Cost: 30.0
```

## A Valid Solution?

The solver chose the arcs depot ↔ A and B ↔ C with a cost of 30.

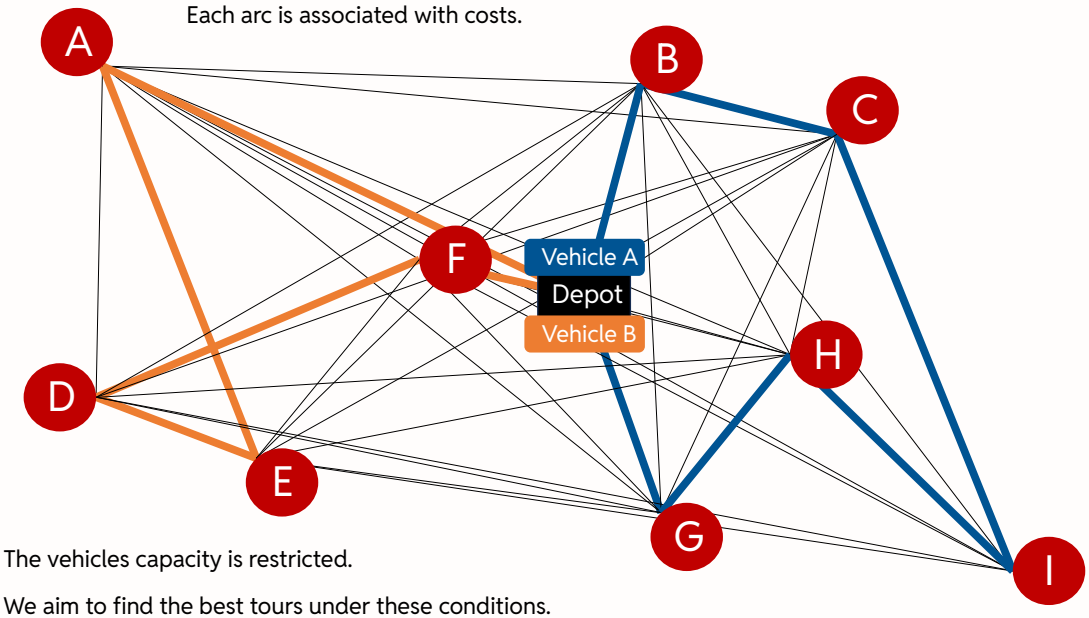
**Question: Is this a valid tour for one vehicle?**

- No! The loop B ↔ C **never visits the depot**
- Our model is still **missing constraints**
- We will fix this in the following sections!

## Problem Constraints

Each customer has a certain demand.

Each arc is associated with costs.




The vehicles capacity is restricted.

We aim to find the best tours under these conditions.

## Constraints?

**Question: What constraints do we need?**

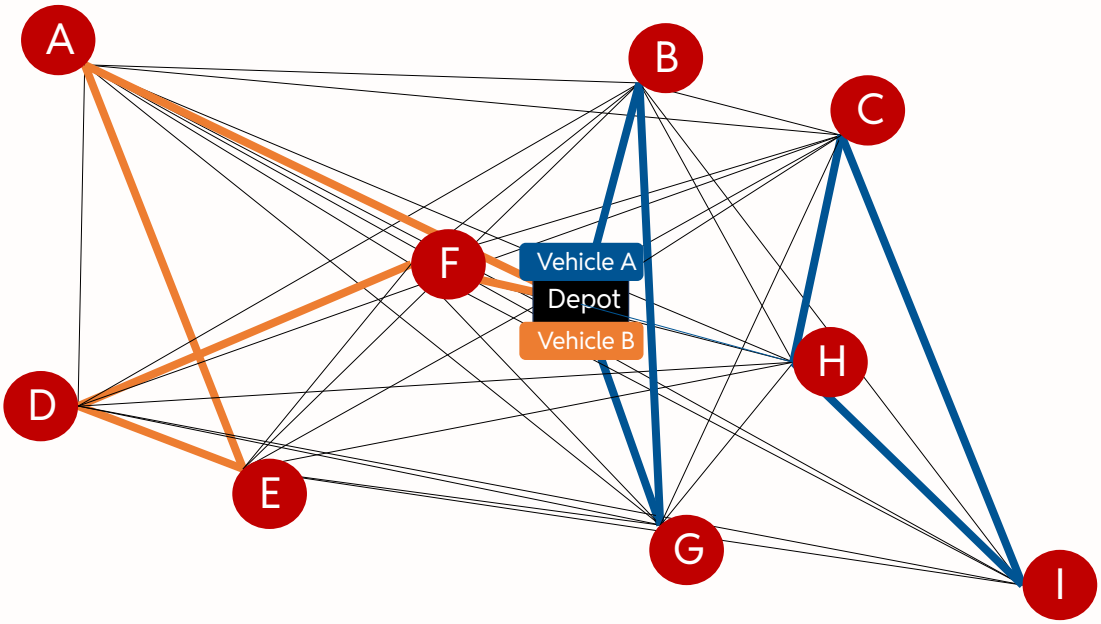
- Each customer has to be **visited once**
- The depot has to be **entered and left** $|\mathcal{K}|$ times
- We have to enforce the **capacity of our vehicles**
- We have to ensure the **maximal duration of each tour**

> ! **Subtours**
> In addition, we have to prevent **subtours**!

What is a

subtour?

## Subtours




# Constraints

## Visit Each Customer Once?

> ! **The goal of these constraints is to:**
> Ensure that each customer is visited exactly once. Essentially, we could also say that each node **has to be entered and left exactly once.**

> **We need the following sets and variables:**
> - $\mathcal{V}$ - Set of all nodes, index $i \in \{0, 1, 2, ..., n\}$
> - $X_{i,j}$ - 1, if the arc between $i$ and $j$ is part of a tour, 0 otherwise

## Visit Each Customer Once

**Question: What could the constraint look like?**

$$\sum_{i \in \mathcal{V}, i \neq j} X_{i,j} = 1 \quad \forall j \in \mathcal{V} \setminus \{0\}$$

$$\sum_{j \in \mathcal{V}, j \neq i} X_{i,j} = 1 \quad \forall i \in \mathcal{V} \setminus \{0\}$$

**Question: Why for all nodes except the depot?**

The depot is the **only node** that is **visited multiple times!**

## Depot Entry/Exit Constraints?

> **The goal of these constraints is to:**
> Ensure that each vehicle enters and leaves the depot exactly $|\mathcal{K}|$ times, as we have $|\mathcal{K}|$ vehicles and each vehicle has to return to the depot.

> **We need the following sets and variables:**
> - $\mathcal{V}$ - Set of all nodes, index $i \in \{0, 1, 2, ..., n\}$
> - $|\mathcal{K}|$ - Number of vehicles
> - $X_{i,j}$ - 1, if the arc between $i$ and $j$ is part of a tour, 0 otherwise

**Question: What could the constraint look like?**

## Depot Entry/Exit Constraints

$$\sum_{i \in \mathcal{V} \setminus \{0\}} X_{i,0} = |\mathcal{K}|$$

$$\sum_{j \in \mathcal{V} \setminus \{0\}} X_{0,j} = |\mathcal{K}|$$

> **Are all constraints necessary?**
> No, theoretically we could also say that we only have to leave **or** enter the depot exactly $|\mathcal{K}|$ times, as the other constraint is already enforced by the "visit each customer once constraint".

## Exactly $|\mathcal{K}|$ Tours?

**Question: What if 3 vehicles would suffice, but we own 5?**

- The constraints **force** all $|\mathcal{K}|$ vehicles to be used

- With generous capacity, this can lead to **more tours than needed** and thus longer total travel times
- Replacing $=$ with $\leq$ would allow the solver to **leave vehicles unused**

# Capacity & Subtours

The next ones are

a little bit tricky.

## MTZ Formulation

- **Miller-Tucker-Zemlin (MTZ)** Constraints from 1960
- Here, in the CVRP form of I. Kara, G. Laporte, and T. Bektas [1]
- Prevent subtours and **track routes** and **capacity utilization**
- First, we need **an additional variable!**

## The MTZ Variable

- $U_i$ - Capacity utilization at $i$ of vehicle on its tour with $i \in \mathcal{V} \setminus \{0\}$
- Note that $U_i$ is a **continuous** variable, not a binary one!

> **Note**
> You don't need to guess these constraints, as they are quite tricky!

## MTZ Constraints

> **We need the following sets, parameters, and variables:**
> - $\mathcal{V}$ - Set of all nodes, index $i \in \{0, 1, 2, ..., n\}$
> - $X_{i,j}$ - 1, if the arc between $i$ and $j$ is part of a tour, 0 otherwise
> - $U_i$ - Capacity utilization at $i$ of vehicle on its tour with $i \in \mathcal{V} \setminus \{0\}$
> - $b$ - Capacity per vehicle (all are identical!)
> - $d_i$ - Demand at node $i$

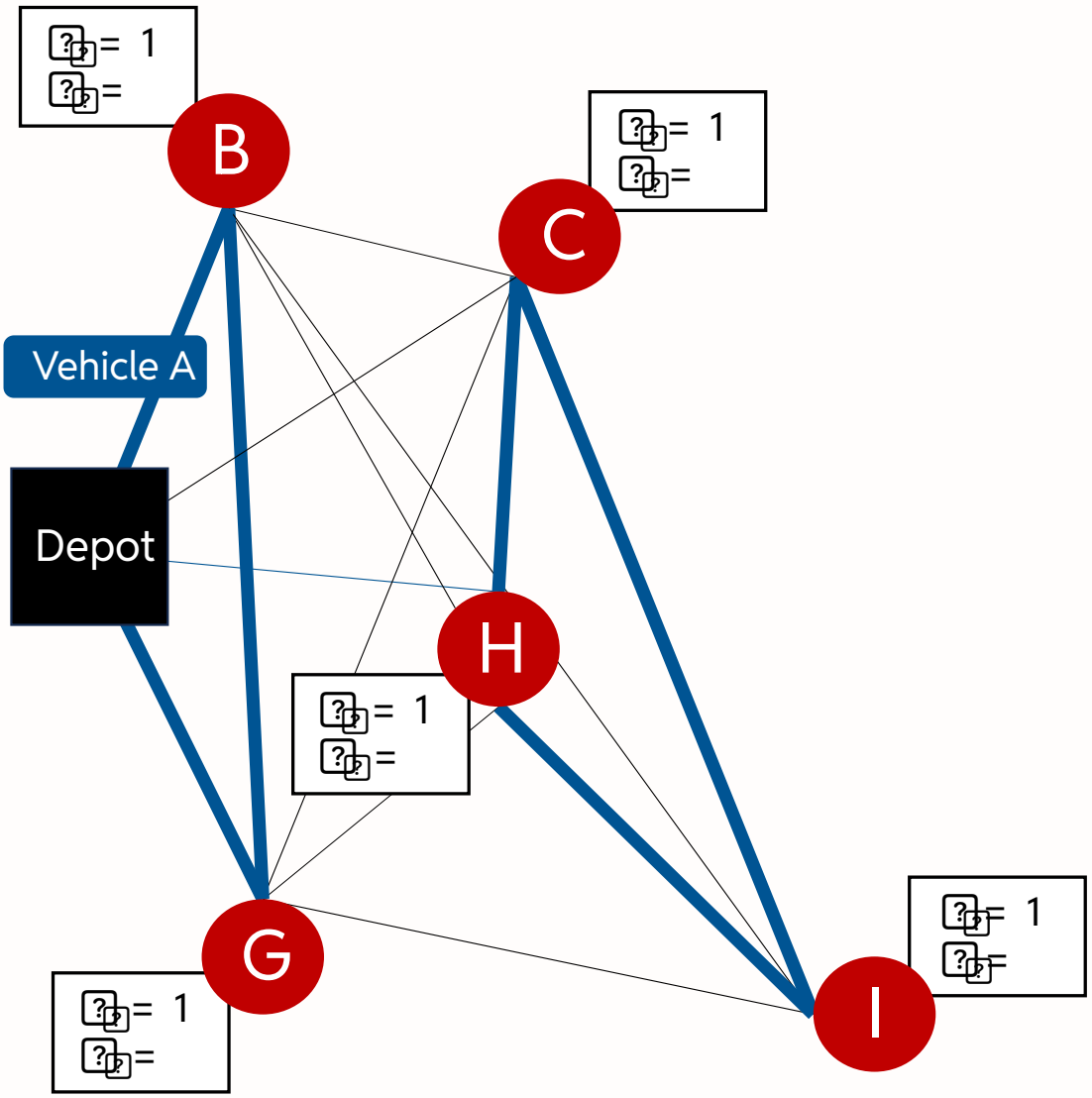
$$U_i - U_j + b \times X_{i,j} \leq b - d_j \quad \forall i, j \in \mathcal{V} \setminus \{0\}, i \neq j$$

$$d_i \leq U_i \leq b \quad \forall i \in \mathcal{V} \setminus \{0\}$$

Too

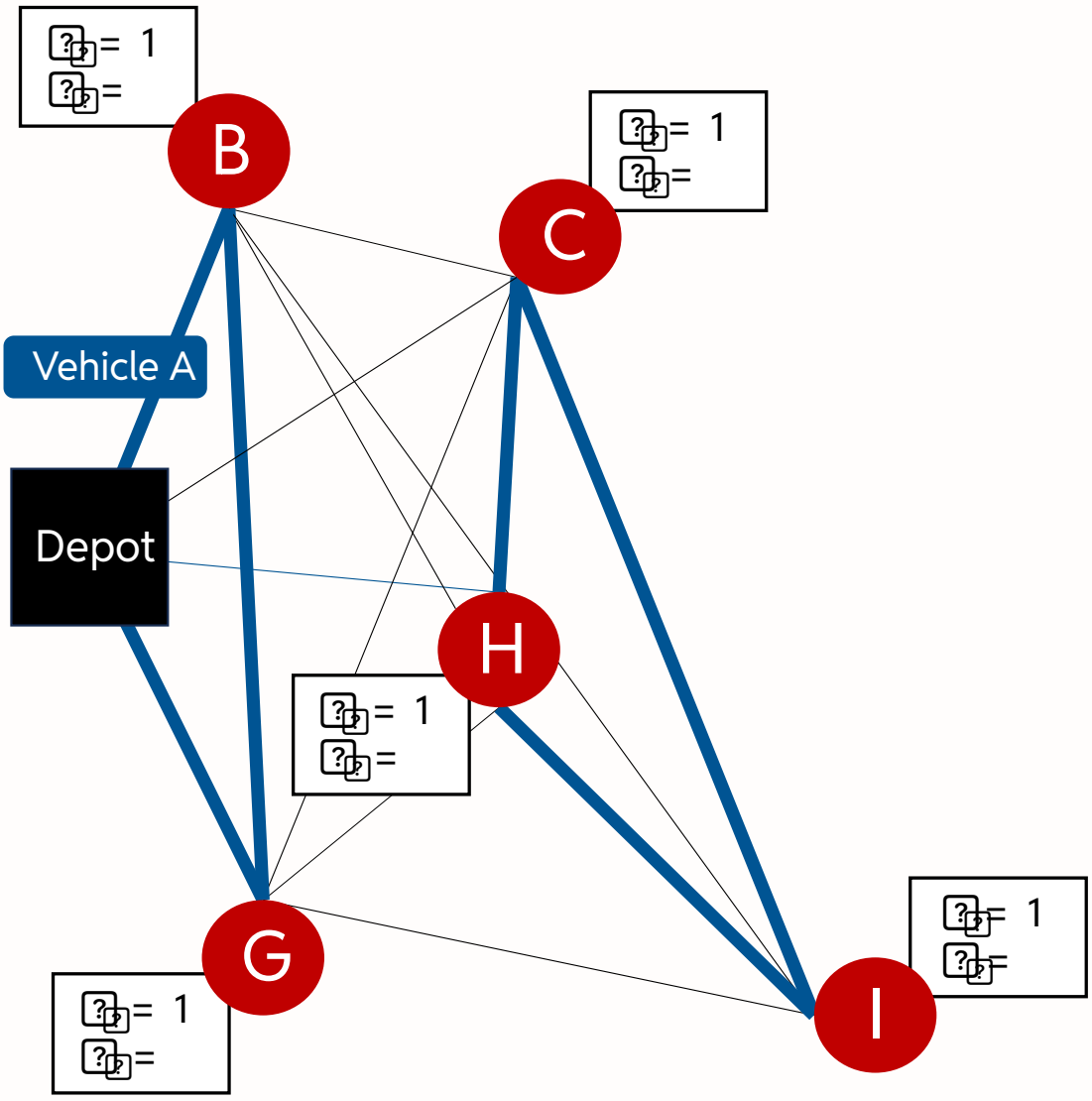
complicated?

## Don't worry!



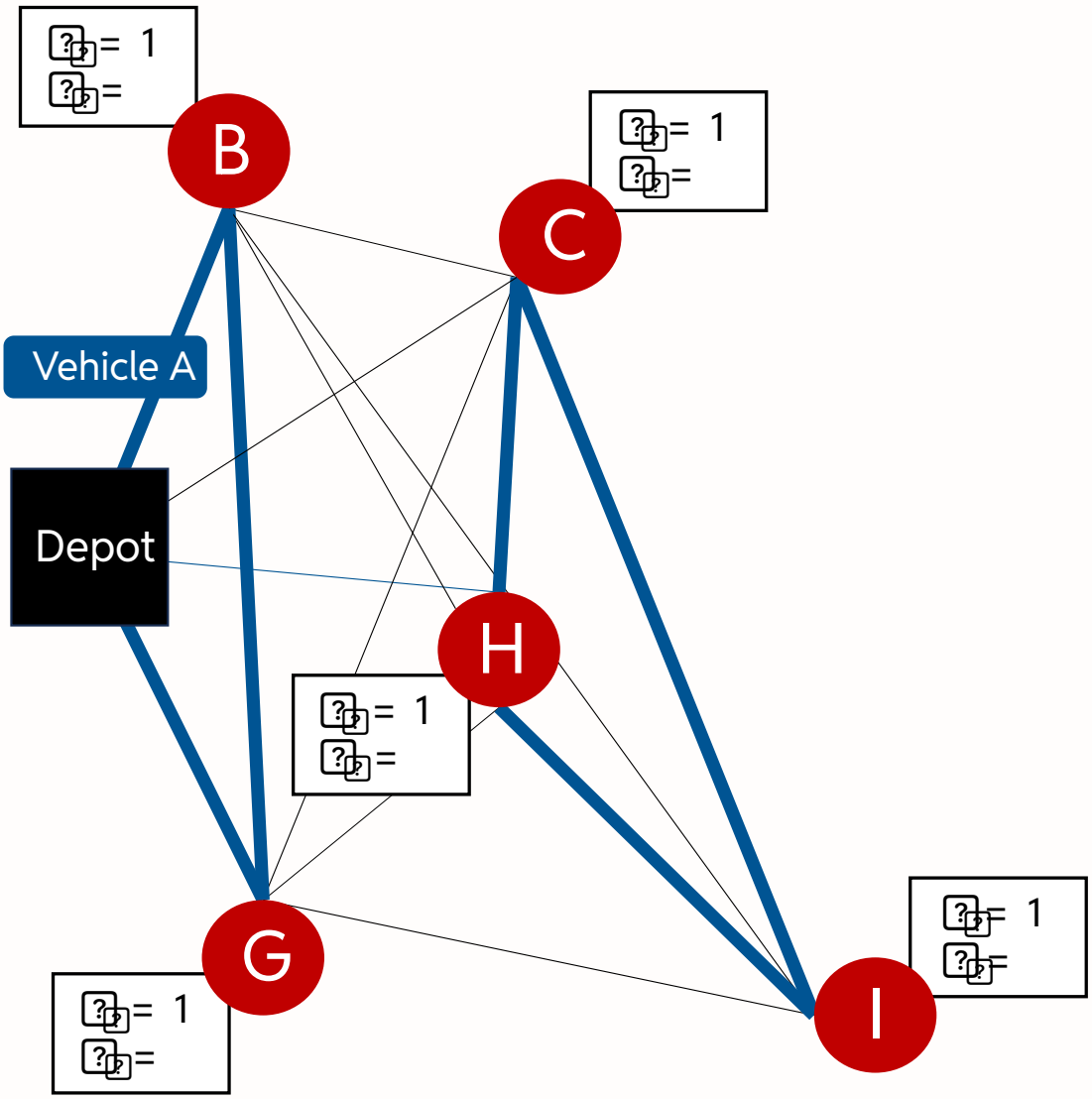

- Let's **break it down!**
- $d_i$ for **all customers** is 1
- Capacity $b$ **per vehicle** is 5
- $U_i$ is the current **capacity utilization** at node $i \in \mathcal{V} \setminus \{0\}$

## No connection between nodes




- In case $X_{i,j} = 0$:
  - $U_i - U_j \leq b - d_j$
  - **Non-binding** for relation between two nodes
- Following is **perfectly fine**:
  - $U_i \leq b$ and $U_j \geq d_j$

## Connection between two nodes




- In case $X_{i,j} = 1$:
  - $U_i - U_j + b \leq b - d_j$
  - **Binding** for relation between two nodes
- **Can be summarized to**:
  - $U_j \geq d_j + U_i$

## Connection in more detail

**Question: Why is it binding?**

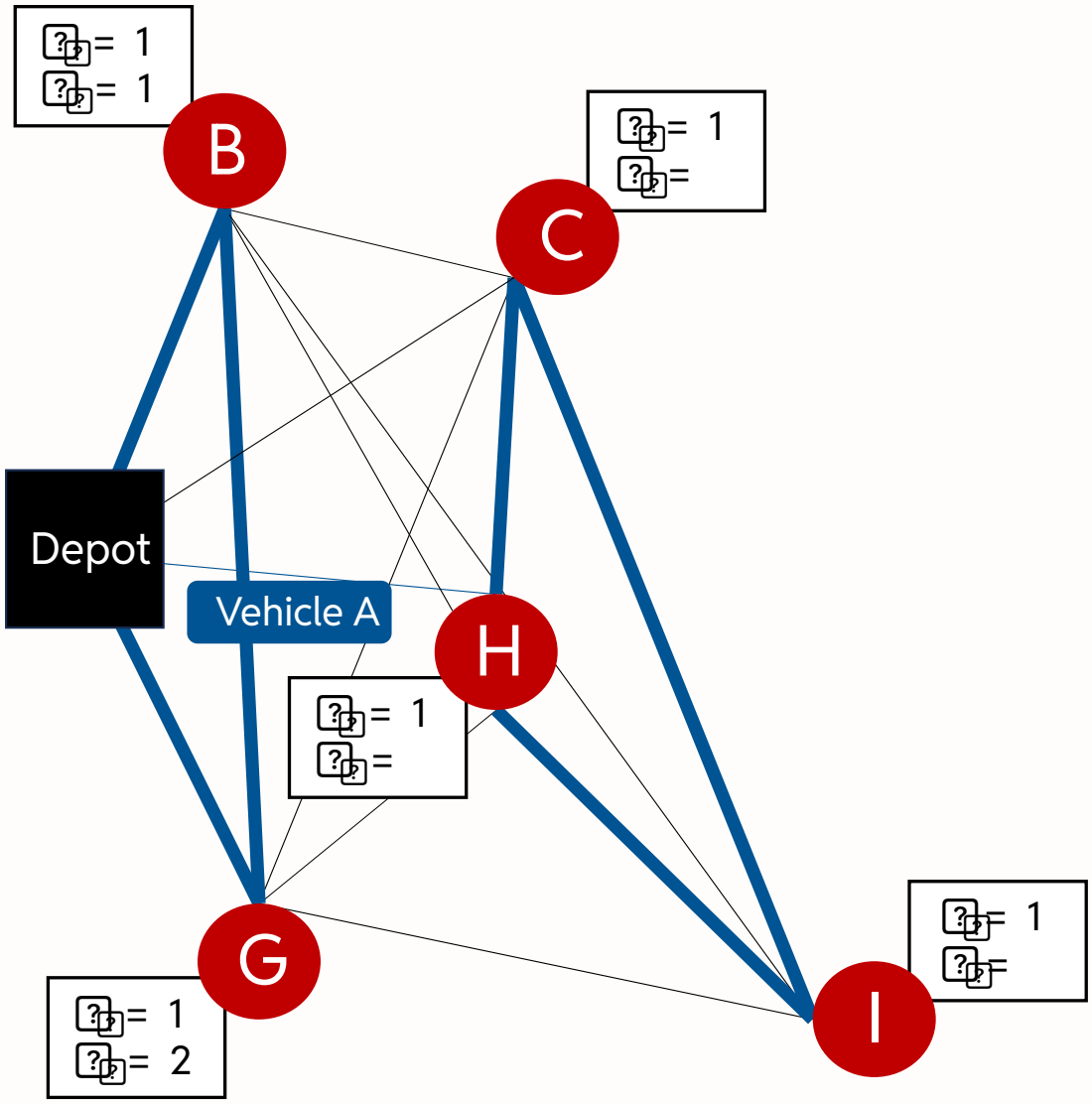
- Binding as $U_j$ has to be **at least as large** as $d_j + U_i$
- Hence, fulfilled if the demand of $j$ is **added to the vehicle**

**Question: Do you get the idea?**

- If $X_{i,j} = 1$, then $U_j$ has to be **at least as large** as $d_j + U_i$

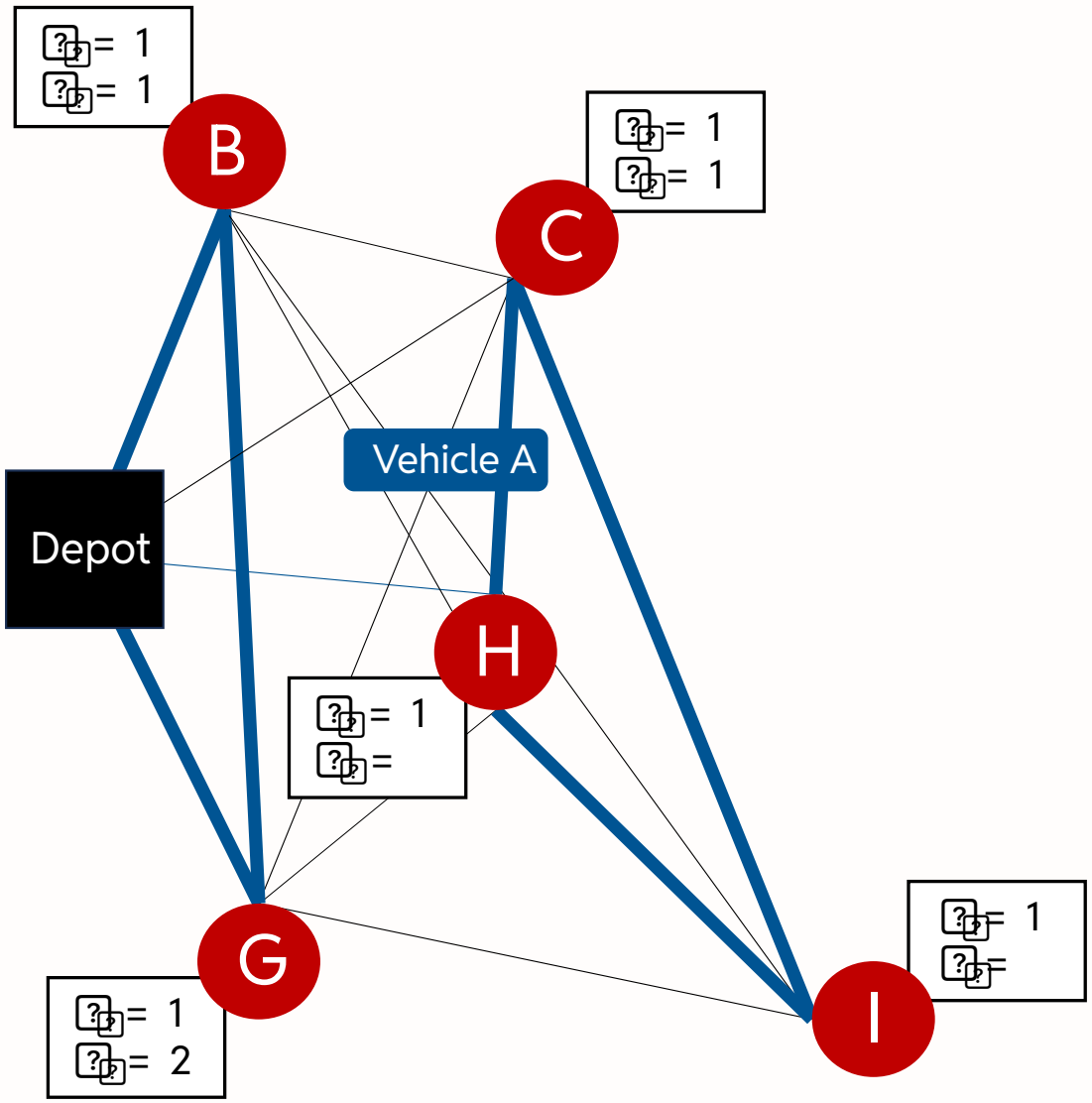
- If $X_{i,j} = 0$, then $U_i \leq b$ and $U_j \geq d_j$

## Tour from the Depot



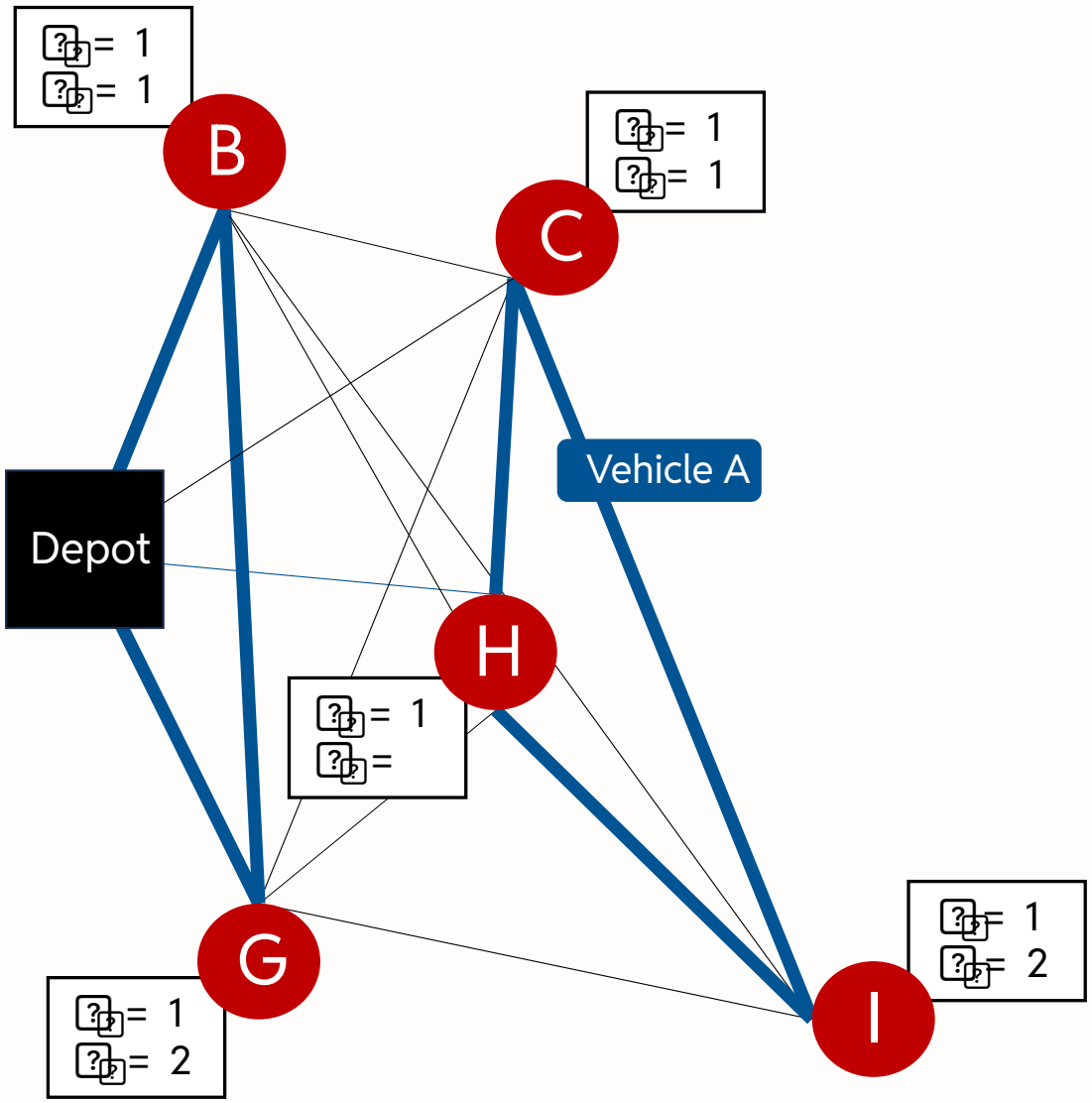

- Tour of vehicle A ok
- Depot is the only node **visited multiple times**
- But the constraints are **not applied here!**

## Tour on its Own



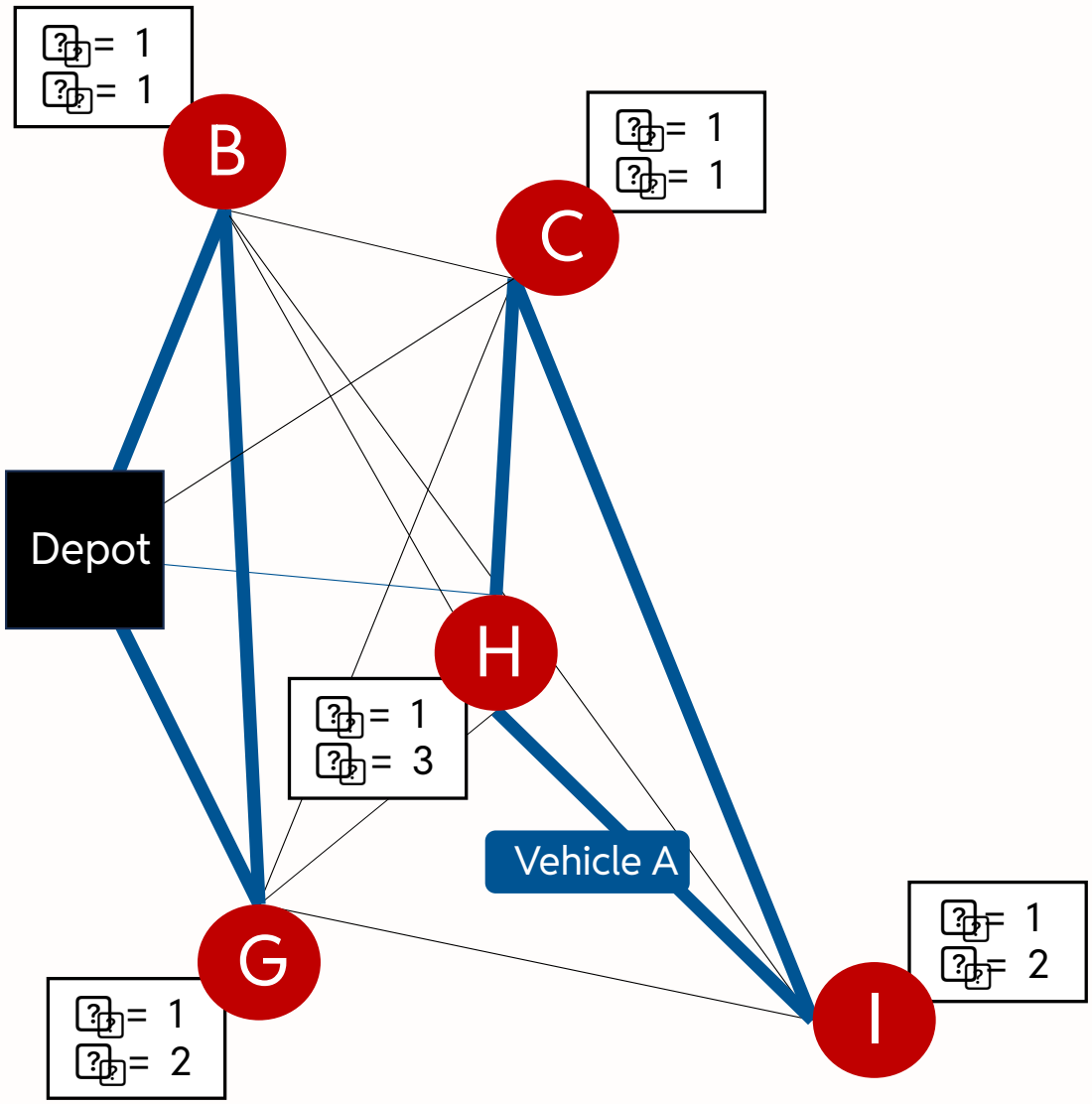

- Assume a subtour $H \to C \to I \to H$
- Each arc on it has $X_{i,j} = 1$
- From $H$ to $C$:
- $U_C \geq U_H + 1$

## Tour on its Own II




- From $C$ to $I$:
- $U_I \geq U_C + 1$
- $U$ has to **increase** along every arc of the subtour

## Tour on its Own III




- From $I$ back to $H$:
- $U_H \geq U_I + 1$
- Now, add up all three constraints
- All $U$ cancel out: $0 \geq 3$
- **Impossible!** No feasible values for $U$ exist!

## Subtour Elimination

- The same argument holds for **any** subtour without the depot
- **Only depot as "reset"**, as constraints are not applied here

**Question: What about the capacity?**

- Remember variable domain of $U_i$?
- $d_i \leq U_i \leq b \rightarrow$ **Overall capacity limit enforced!**

# Last Constraint

## Ensure time limit?

**Question: Does anyone have an idea?**

- Constraints basically **follow the same idea**!
- First, we again need an additional variable
- $T_i$ - Time spent on tour at node $i$ of a vehicle with $i \in \mathcal{V} \setminus \{0\}$
- $T_i$ is a **continuous** variable, just like $U_i$

## Ensure time limit

**We need the following sets and variables:**

- $\mathcal{V}$ - Set of all nodes, index $i \in \{0, 1, 2, ..., n\}$
- $X_{i,j}$ - 1, if the arc between $i$ and $j$ is part of a tour, 0 otherwise
- $T_i$ - Time spent on tour at the node $i$ of a vehicle with $i \in \mathcal{V} \setminus \{0\}$
- $t$ - Maximal duration of a tour
- $c_{i,j}$ - Travel time on an arc from $i$ to $j$

$$T_i - T_j + t \times X_{i,j} \leq t - c_{i,j} \quad \forall i, j \in \mathcal{V} \setminus \{0\}, i \neq j$$

$$c_{0,i} \leq T_i \leq t - c_{i,0} \quad \forall i \in \mathcal{V} \setminus \{0\}$$

## Time limit and the depot

**Question: Why do we need the tightened domain of $T_i$?**

- $T_i \geq c_{0,i}$: the drive **from the depot** to the first customer counts
- $T_i \leq t - c_{i,0}$: time to **return to the depot** is reserved
- With just $0 \leq T_i \leq t$, a tour could **exceed the limit** by the two depot legs!

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Any questions?

# Asymmetric CVRP

## Objective

$$\text{Minimize} \quad \sum_{(i,j) \in \mathcal{A}} c_{i,j} \times X_{i,j}$$

**The goal of the objective function is to:**
Minimize the total travel distance.

## Each customer is visited once

$$\sum_{i \in \mathcal{V}, i \neq j} X_{i,j} = 1 \quad \forall j \in \mathcal{V} \setminus \{0\}$$

$$\sum_{j \in \mathcal{V}, j \neq i} X_{i,j} = 1 \quad \forall i \in \mathcal{V} \setminus \{0\}$$

! **Our constraints ensure:**
Each customer is visited exactly once.

## Depot entry and exit

$$\sum_{i \in \mathcal{V} \setminus \{0\}} X_{i,0} = |\mathcal{K}|$$

$$\sum_{j \in \mathcal{V} \setminus \{0\}} X_{0,j} = |\mathcal{K}|$$

! **Our constraints ensure:**
The depot is visited by exactly $|\mathcal{K}|$ vehicles. Note, that we could remove one of the constraints and the solution would still be optimal.

## Capacity/subtour elimination

$$U_i - U_j + b \times X_{i,j} \leq b - d_j \quad \forall i, j \in \mathcal{V} \setminus \{0\}, i \neq j$$

$$d_i \leq U_i \leq b \quad \forall i \in \mathcal{V} \setminus \{0\}$$

! **Our constraints ensure:**
The capacity limit is respected and subtours are eliminated.

## Time constraints

$$T_i - T_j + t \times X_{i,j} \leq t - c_{i,j} \quad \forall i, j \in \mathcal{V} \setminus \{0\}, i \neq j$$

$$c_{0,i} \leq T_i \leq t - c_{i,0} \quad \forall i \in \mathcal{V} \setminus \{0\}$$

! **Our constraints ensure:**
The time limit is respected (and subtours are eliminated). The domain of $T_i$ accounts for the travel from and back to the depot.

## Variables

$$X_{i,j} \in \{0, 1\} \quad \forall (i, j) \in \mathcal{A}$$

$$d_i \leq U_i \leq b \quad \forall i \in \mathcal{V} \setminus \{0\}$$

$$c_{0,i} \leq T_i \leq t - c_{i,0} \quad \forall i \in \mathcal{V} \setminus \{0\}$$

> ! **The variable domains make sure that:**
> The binary setup variable is either 0 or 1 and the continuous variables $U_i$ and $T_i$ stay within the capacity and time limits.

# Model Characteristics

## Characteristics

**Questions: On model characteristics**

- Is the model formulation linear/ non-linear?
- What kind of variable domains do we have?
- What do you think, can the model be solved quickly?

## Model Assumptions

**Questions: On model assumptions**

- What assumptions have we made?
- What are issues that can arise if the model is applied?
- Have we considered service times?

## Model Limitations

Our formulation makes **several simplifying assumptions**:

- **No service time** at customer locations
- **Homogeneous fleet** (all vehicles are identical)
- **Deterministic demand** (known exactly in advance)
- **Single depot** (all vehicles start and end at same location)

## Real-World Complications

### Operational Challenges

- Traffic variability
- Driver breaks and regulations
- Vehicle breakdowns
- Weather conditions

### Planning Challenges

- Dynamic customer requests
- Uncertain demand
- Last-minute cancellations
- Multi-day horizons

> **Tip**
> These complications often require **extensions** of the basic CVRP model or **robust optimization** approaches!

### Extensions of the CVRP

**Questions: What extensions do you know?**

- time windows (TW)
- soft time windows (STW)
- multiple depots (MD)
- heterogeneous fleet (HF)
- backhauls (B)
- pickup and delivery (PD)

# In Practice

## Case Study in Schleswig-Holstein

- 165 libraries
- 119 visited biweekly
- Up to 9 different tours
- Time-Limit of 8 hours for each tour

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Can our formulation
solve the problem
on a real-world instance?

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No, although we can find
solutions within one hour,
the gap is still very large
with 40%-45%.

### Problem is NP-hard

- We have already encountered **NP-hard problems** in this course
- Likely, that there are **no polynomial-time algorithms**

- **Doesn't mean that it can't be solved!**

> **Note**
> The CVRP is a generalization of the **Traveling Salesman Problem (TSP)**, which is itself NP-hard!

## Why Does This Matter?

The solution space grows **explosively** with problem size:

| Nodes | Possible Tour Orders | Approx. Computation |
|---|---|---|
| 10 | 3.6 million | Seconds |
| 15 | 1.3 trillion | Minutes |
| 20 | 2.4 × 10^18 | Hours to days |
| 50 | 3 × 10^64 | Infeasible |

**Question: Why is the 40-45% gap after 1 hour expected?**

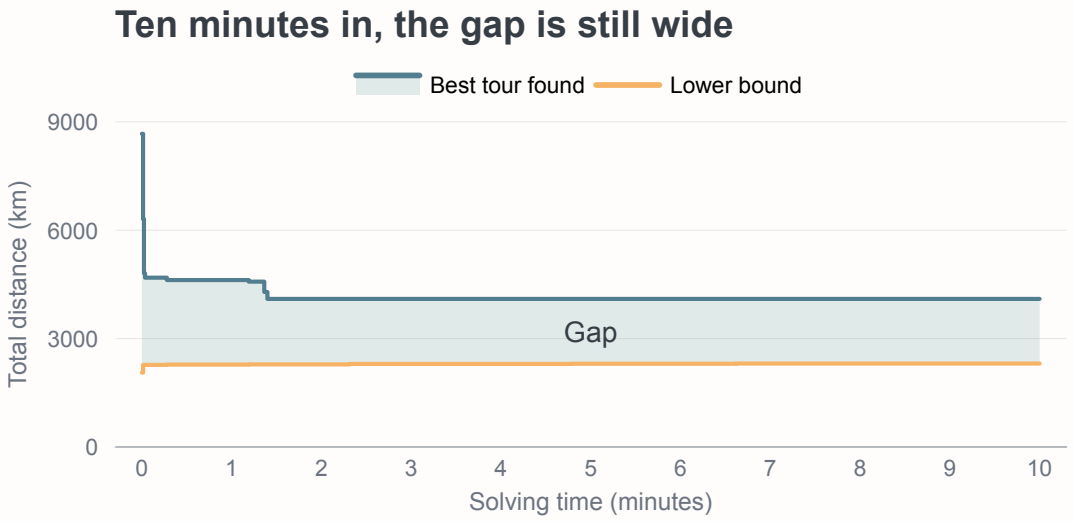
## Understanding Optimality Gap

- The **optimality gap** measures:

$$\frac{\text{Best Solution} - \text{Lower Bound}}{\text{Lower Bound}} \times 100\%$$

- A 40% gap means our solution **could be** up to 40% worse than optimal[2]

## The Gap Over Time




- 45 random customers, solved with our model in HiGHS

[2] Note that most solvers (including HiGHS) report the gap relative to the best solution, $\frac{\text{Best Solution} - \text{Lower Bound}}{\text{Best Solution}}$, so the numbers in the solver log differ slightly from our definition.

- The best tour improves in jumps, the **lower bound hardly moves**

## Consequences of the Gap

- For 165 libraries, even **state-of-the-art solvers** struggle
- **This is why we need heuristics!**

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Can we do

anything to solve

the model?

## Heuristics

- We can still solve the problem **with a heuristic**
- **Likely not the optimal solution**, but a lot of research goes into efficient algorithms to solve these problems

> **Tip**
> For problems with 100+ locations, heuristics are often the only practical choice.

## Heuristics for VRP

**Construction**

Build a solution from scratch

- Nearest neighbor
- Savings algorithm
- Sweep method

**Improvement**

Refine existing solutions

- 2-opt (swap edges)
- Or-opt (relocate)
- Exchange moves

**Metaheuristics**

Intelligent search strategies

- Genetic algorithms
- Simulated annealing
- Tabu search

> **Tip**
> Interested in more details? Check the lecture Management Science

## HGS-CVRP: State-of-the-Art

In our case study we applied **Hybrid Genetic Search** for the CVRP (HGS-CVRP) by T. Vidal [2]

- Maintains **diverse population** of solutions
- Applies **intensive local search** to improve offspring
- Achieves near-optimal solutions in **seconds to minutes**

> **Tip**
> For real-world applications, also consider OR-Tools by Google, it is open-source and production-ready!

## Case Study Results

In our project, using **HGS-CVRP** instead of exact optimization:

- Solution found in **< 5 minutes** (vs. hours with MIP)
- Total driving distance reduced by **~20%** compared to manual planning
- $CO_2$ emissions reduced by approximately **12-15 tonnes/year**

> **Important**
> The heuristic solution is **better** than what the MIP solver found in 1 hour!

## Applications Beyond Libraries

**Logistics & Delivery**

- Package delivery
- Food delivery services
- Grocery delivery
- Mail distribution

**Service Industries**

- Maintenance crews
- Home healthcare visits
- Waste collection
- School bus routing

**Question: Can you think of other applications?**

## Conclusion

- Standard problem that **occurs in many different places**
- Solving the problem with a mathematical model **is difficult**
- Nowadays, there are **many good heuristics**
- **Many companies** are working on the problem

> **And that's it for today's lecture!**
> We now have covered the Capacitated Vehicle Routing Problem and are ready to start solving some tasks in the upcoming tutorial.

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Questions?

# Literature

For more interesting literature to learn more about Julia, take a look at the literature list of this course.

[1] I. Kara, G. Laporte, and T. Bektas, "A note on the lifted Miller–Tucker–Zemlin subtour elimination constraints for the capacitated vehicle routing problem," *European Journal of Operational Research*, vol. 158, no. 3, pp. 793–795, 2004, doi: https://doi.org/10.1016/S0377-2217(03)00377-1.

[2] T. Vidal, "Hybrid genetic search for the CVRP: Open-source implementation and SWAP* neighborhood," *Computers & Operations Research*, vol. 140, p. 105643, Apr. 2022, doi: 10.1016/j.cor.2021.105643.